DECODELABS

Exploratory Data Analysis

Submission Report — Project 2

E-Commerce Order Dataset (1,200 records, Jan 2023 – Jun 2025)

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1. Problem Statement

DecodeLabs operates an e-commerce platform selling seven product categories (Phones, Laptops, Tablets, Monitors, Printers, Chairs, Desks). The business needs a clear, evidence-based picture of order behavior — average order value, revenue trends, category performance, and the drivers behind order value — before any predictive modeling or dashboarding work begins.

This report performs an Exploratory Data Analysis (EDA) on the raw order dataset to answer: What is the typical order worth, which factors most influence it, which orders represent unusual but legitimate activity, and what should the business act on?

2. Methodology

The analysis followed the IPO framework (Input → Process → Output):

- **Input: 1,200 order records across 14 fields, covering order details, customer ID, product, pricing, fulfillment status, and marketing attribution.**
- Process: Data quality audit → descriptive statistics (mean, median, std, quartiles) → univariate distribution analysis → outlier detection (IQR method) → categorical comparisons → correlation analysis.
- Output: This report plus an accompanying Excel workbook (EDA_Analysis_Workbook.xlsx) containing the raw data, statistics tables, and pivoted summaries.

Tools used: Python (pandas, matplotlib) for computation and visualization; Microsoft Excel for the structured data deliverable.

3. Data Quality Audit

Before any statistical interpretation, the dataset was checked for completeness and structural integrity (the “forensic” step recommended in the training material).

Check	Result
Total records	1,200
Total columns	14
Fully complete columns	13 of 14
Column with missing values	CouponCode — 309 blanks (25.8%)
Interpretation of missing values	Not an error — a blank CouponCode simply means the customer checked out without applying a coupon. No imputation needed.
Duplicate Order IDs	None found
Date range	Jan 1, 2023 – Jun 30, 2025

Conclusion: the dataset is clean and analysis-ready. The only “missing” field is a legitimate business state (no coupon used), not a data-entry fault.

4. Descriptive Statistics

The five-number summary and central-tendency measures were calculated for all numeric fields:

Metric	Quantity	Unit Price ($)	Items In Cart	Total Price ($)
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std. Dev.	1.41	197.18	2.28	819.86
Minimum	1	11.39	1	11.39
Q1 (25th pct)	2	186.06	4	410.52
Q3 (75th pct)	4	521.57	7	1,578.48
Maximum	5	699.93	10	3,456.40
Skewness	≈0 (symmetric)	≈0 (symmetric)	≈0 (symmetric)	0.89 (right-skewed)

Key observation: Total Price has a mean ($1,053.97) noticeably higher than its median ($823.62), and a skewness of 0.89 , a classic right-skewed distribution. This means a small number of high-value orders pull the average upward. For reporting typical order value, the median ($823.62) is the more reliable, outlier-resistant figure; the mean is still useful for total revenue forecasting.

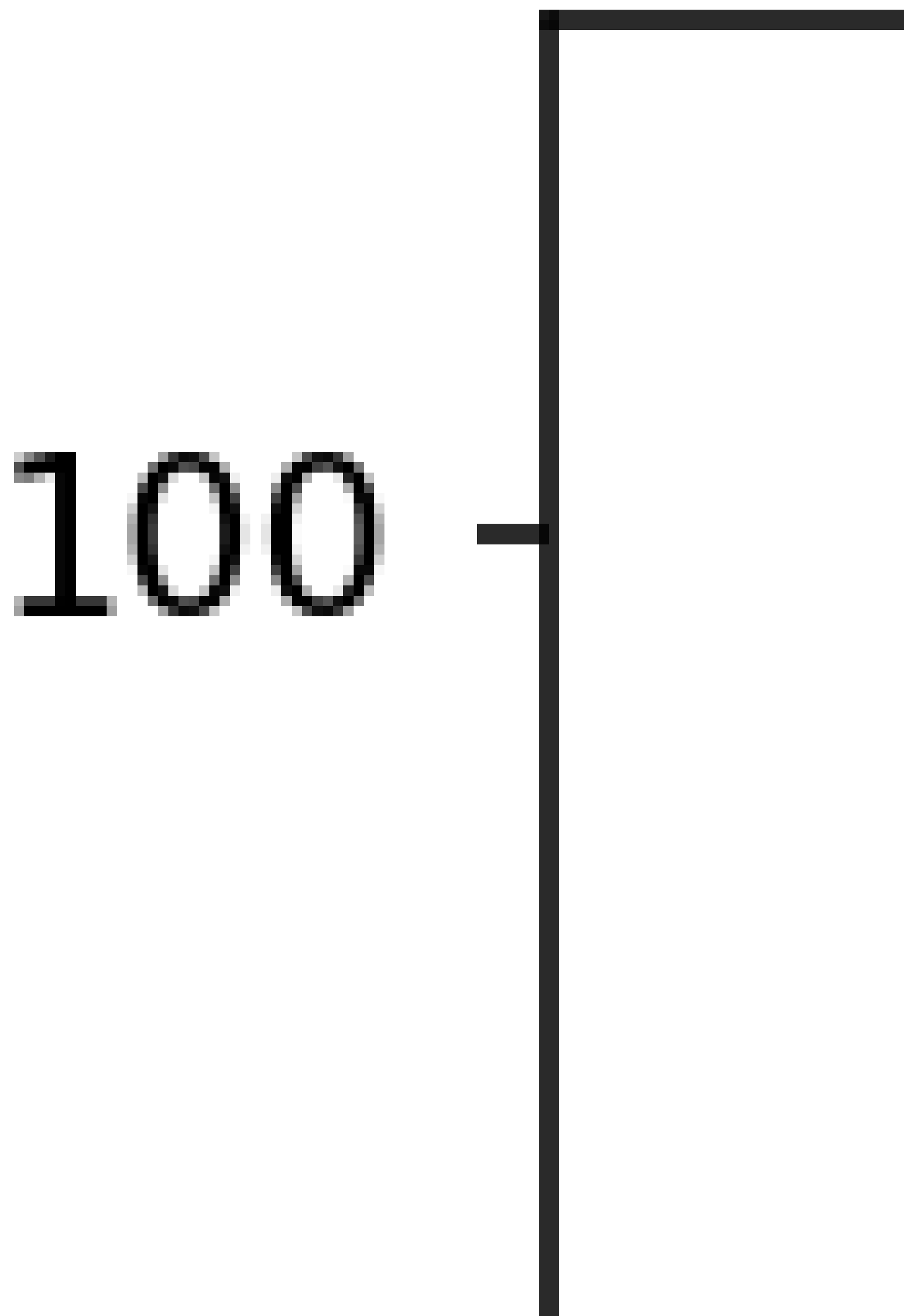
100

Figure 1: Distribution of Order Total Price — right-skewed, confirming mean > median.

5. Outlier Detection

Outliers in Total Price were flagged using the IQR method (the industry-standard approach for skewed, “dirty” business data per the training framework):

- IQR = Q3 − Q1 = $1,578.48 − $410.52 = $1,167.96
- Lower bound = Q1 − 1.5×IQR = −$1,341.41 (no orders below $0, so this bound flags nothing)
- Upper bound = Q3 + 1.5×IQR = $3,330.41
- Result: 8 orders (0.7% of all orders) exceed the upper bound.

Order ID	Product	Qty	Unit Price ($)	Total Price ($)	Status
ORD200789	Tablet	5	691.28	3,456.40	Delivered
ORD201122	Monitor	5	678.19	3,390.95	Returned
ORD200632	Laptop	5	678.16	3,390.80	Delivered
ORD200469	Chair	5	676.98	3,384.90	Cancelled
ORD200328	Tablet	5	674.04	3,370.20	Cancelled
ORD200107	Printer	5	670.75	3,353.75	Shipped
ORD200326	Laptop	5	670.48	3,352.40	Returned
ORD201065	Printer	5	666.80	3,334.00	Delivered

Interpretation — Noise vs. Signal: every flagged order is a maximum-quantity (5-unit) purchase of a near-top-of-range unit price item. There is no formatting corruption, negative value, or impossible figure among them. These are classified as SIGNAL (legitimate bulk/high-ticket purchases), not data errors, and should be investigated as a high-value customer segment rather than cleaned or removed.

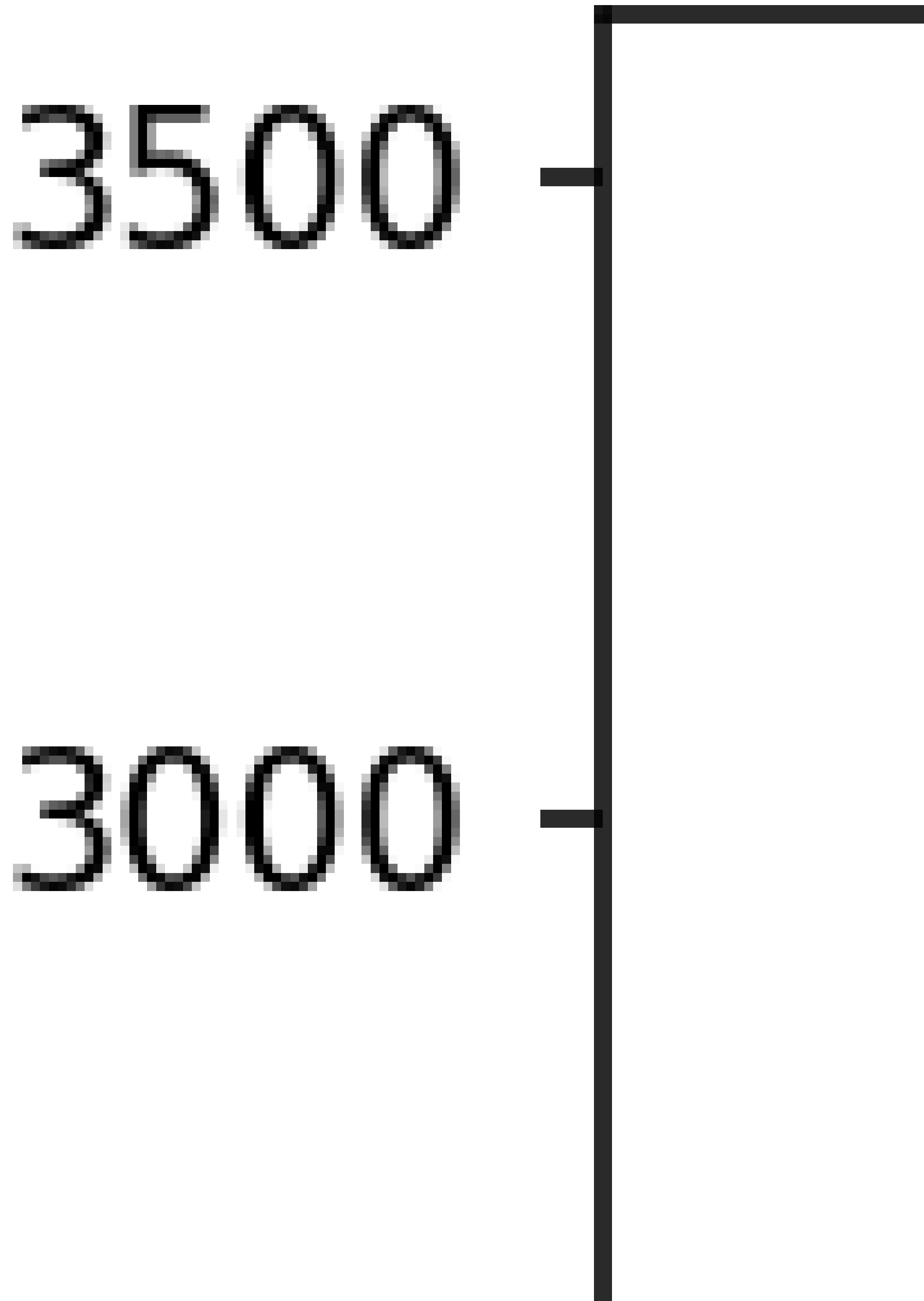
3500
3000

Figure 2: Order value spread by product — outlier points visible above each whisker.

6. Category and Channel Performance

6.1 Revenue by Product

Product	Orders	Total Revenue ($)	Avg Order Value ($)
Chair	178	195,620.11	1,098.99
Printer	181	195,612.61	1,080.73
Laptop	173	192,126.56	1,110.56
Tablet	179	186,568.95	1,042.28
Monitor	163	175,651.41	1,077.62
Desk	170	167,459.93	985.06
Phone	156	151,722.39	972.58

Chairs and Printers lead in total revenue despite Laptops having the highest average order value — order volume, not just price, drives category revenue. Phones generate the least revenue, both in volume and average order size.

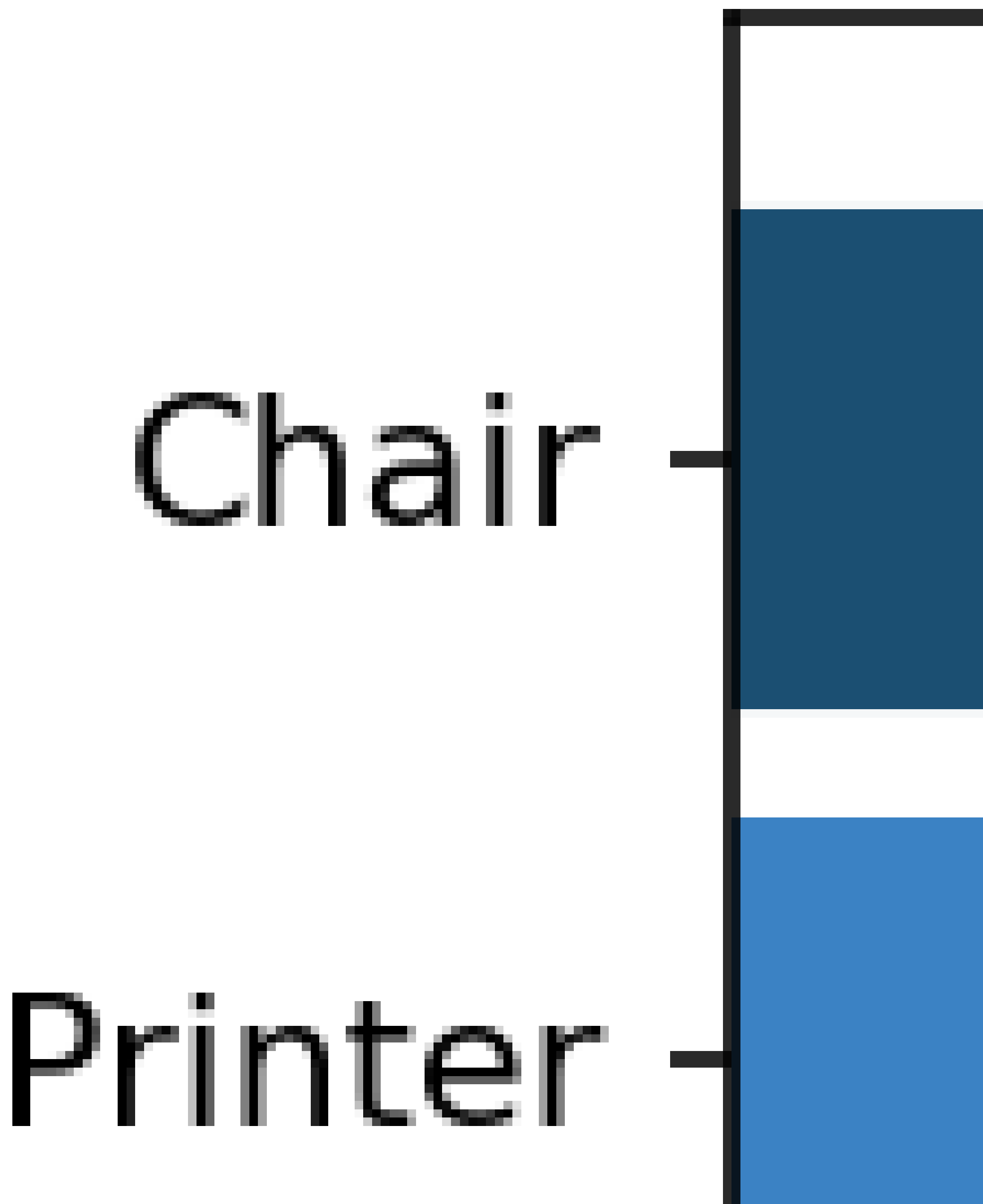
Chair
Printer

Figure 3: Total revenue by product category, sorted.

6.2 Order Status Breakdown

Order Status	Orders	Share of Total
Cancelled	250	20.8%
Returned	247	20.6%
Pending	237	19.8%
Shipped	235	19.6%
Delivered	231	19.3%

Order status is almost evenly split across five states, with Cancelled and Returned combined accounting for 41.4% of all orders. This is a meaningful operational signal worth flagging to fulfillment and quality teams — nearly 2 in 5 orders never convert into a completed sale.

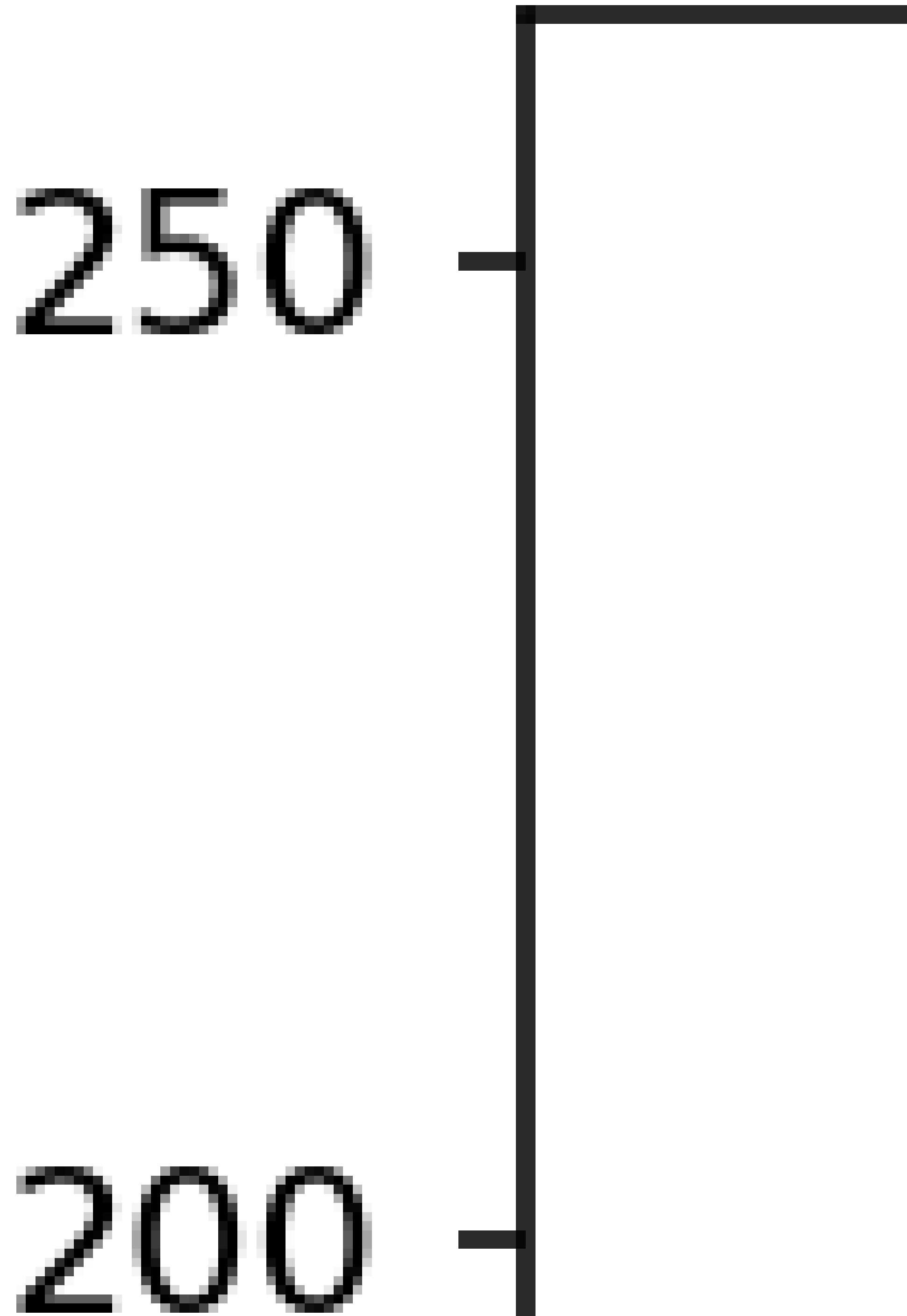
250
200

Figure 4: Order count by fulfillment status.

6.3 Monthly Revenue Trend

Revenue was aggregated by month across the full 30-month window. The trend shows seasonal fluctuation with no single sustained growth or decline trajectory; the most recent month (June 2025) recorded the highest single-month revenue in the dataset ($53,047.40).

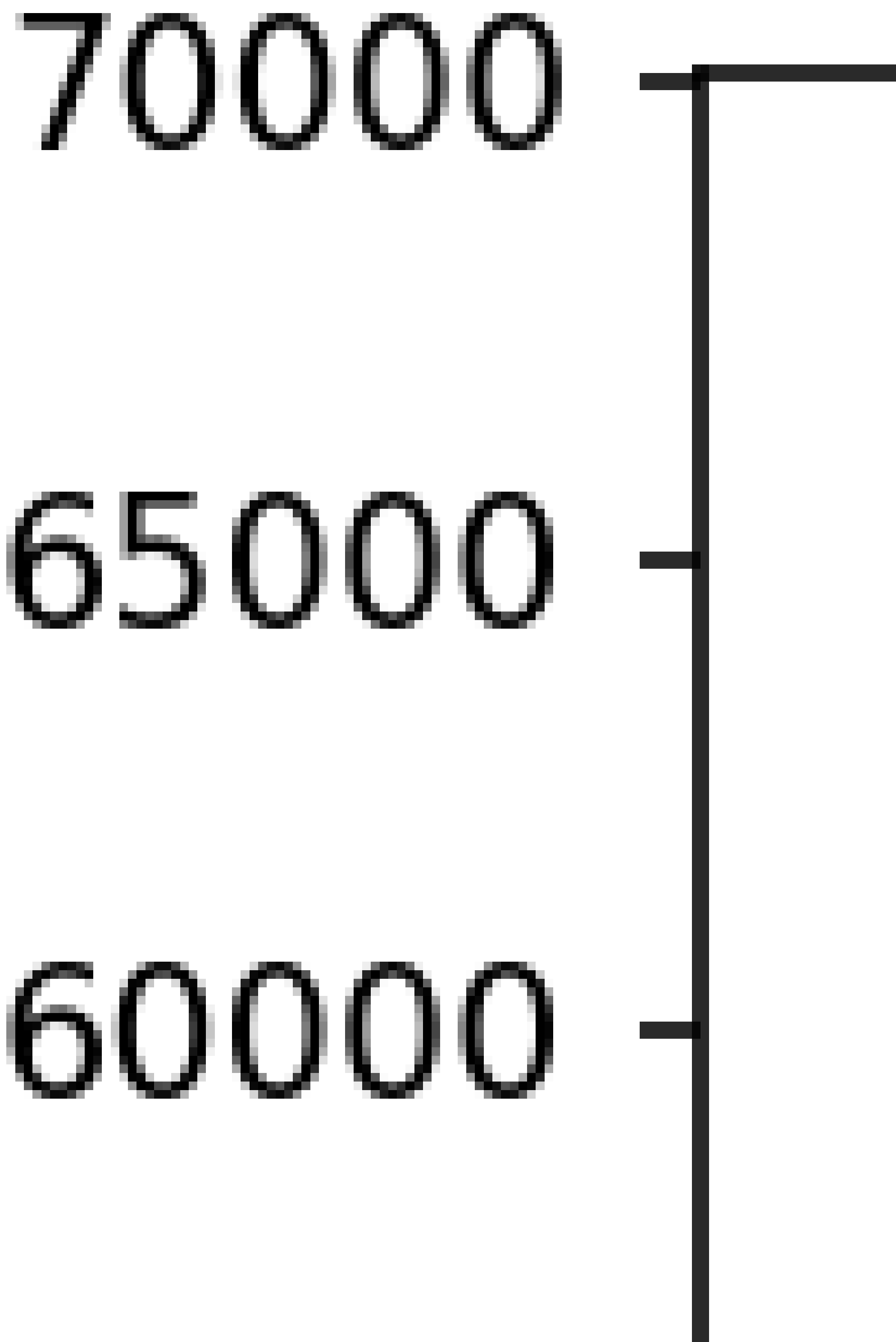
70000
65000
60000

Figure 5: Monthly revenue, January 2023 – June 2025.

7. Correlation Analysis

Pearson correlation coefficients were calculated between the four numeric fields to identify which variables most strongly drive Total Price:

	Quantity	Unit Price	Items In Cart	Total Price
Quantity	1.00	0.01	0.65	0.62
Unit Price	0.01	1.00	0.00	0.72
Items In Cart	0.65	0.00	1.00	0.39
Total Price	0.62	0.72	0.39	1.00

Quantity

Figure 6: Correlation matrix heatmap.

Unit Price (r = 0.72) is the strongest driver of Total Price, followed closely by Quantity (r = 0.62). Items In Cart correlates moderately with Quantity (r = 0.65) — logical, since both reflect basket size — but only weakly with Total Price (r = 0.39), indicating cart size alone is a poor predictor of order value.

Golden Rule reminder: these are correlations, not causal proof. Unit Price driving Total Price is a near-mathematical relationship (Total = Quantity × Unit Price), so this finding confirms data consistency rather than revealing a hidden business insight.

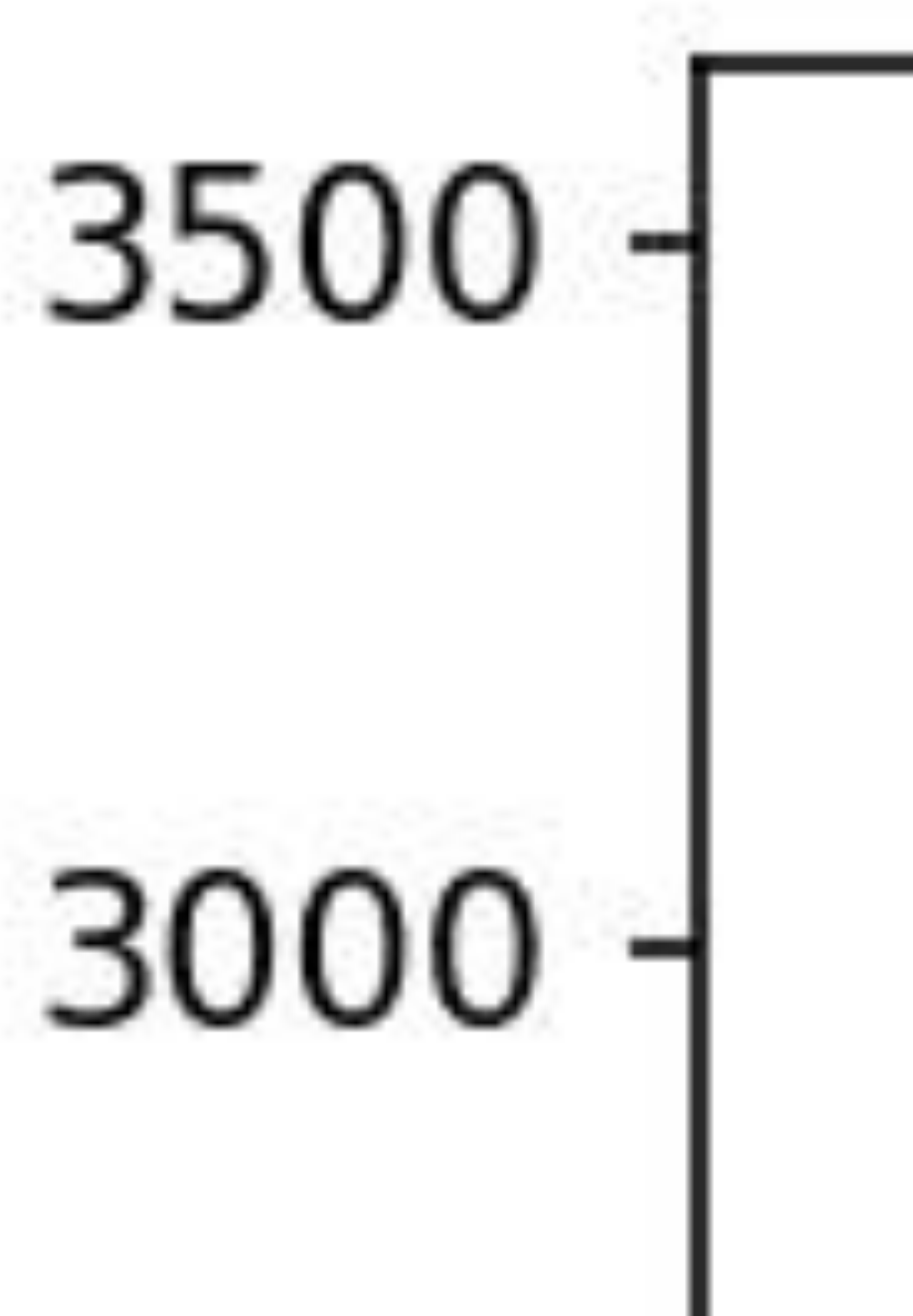
3500
3000

Figure 7: Unit Price vs. Total Price scatter plot — positive linear relationship.

8. Key Findings Summary

- Order value is right-skewed (skew = 0.89): median ($823.62) is a more representative “typical order” figure than the mean ($1,053.97).
- 8 orders (0.7%) are statistical outliers by the IQR method, but all represent legitimate maximum-quantity, high-ticket purchases — a high-value segment worth targeted marketing, not data cleanup.
- Chairs and Printers generate the most total revenue; Laptops have the highest average order value; Phones underperform on both volume and value.
- 41.4% of orders end in Cancelled or Returned status — the single largest opportunity area in the dataset.
- Unit Price (r=0.72) and Quantity (r=0.62) are the two strongest numeric drivers of order value; cart size alone is a weak predictor (r=0.39).
- Revenue shows monthly fluctuation with the latest month (Jun 2025) at a dataset-high of $53,047.

9. Business Recommendations

- **Investigate the Cancelled/Returned segment (41.4% of orders) by payment method and product to identify root causes — this is the highest-leverage fix in the dataset.**
- Build a loyalty or bulk-order program targeting the 8 high-value outlier customers and similar 5-unit purchase patterns identified in this analysis.
- Reassess Phone category pricing/promotion — lowest revenue and lowest average order value of all 7 product lines.
- Use median, not mean, when reporting “average order value” to stakeholders, given the confirmed right skew.
- Extend this analysis with a time-series breakdown by product category to test whether the Jun 2025 revenue spike is seasonal or a genuine growth trend.

10. Conclusion

This EDA transformed 1,200 raw order records into a verified set of business insights: a clean dataset with one explainable gap, a right-skewed order-value distribution, eight legitimate high-value outliers, clear category and fulfillment-status patterns, and quantified relationships between order attributes. These findings establish the diagnostic foundation required before any dashboarding or predictive modeling work begins on this dataset.

Supporting workbook: EDA_Analysis_Workbook.xlsx (raw data, data quality table, descriptive statistics, outlier list, category summaries, monthly trend, and correlation matrix).